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SEMICONDUCTOR PACKAGE INCLUDING AN ENCAPSULANT

Abstract

A semiconductor package includes: a lower redistribution structure including a lower insulating layer and a lower redistribution layer; a semiconductor chip disposed on the lower redistribution structure; connection conductors connected to the lower redistribution layer; an encapsulant disposed on the connection conductors; and an upper redistribution structure including an upper insulating layer and upper redistribution layers, wherein the upper insulating layer is disposed on the encapsulant, wherein the upper redistribution layers are disposed on the upper insulating layer, wherein the connection conductors and the encapsulant form a first step, wherein the upper redistribution layers include first and second upper redistribution layers, wherein the first upper redistribution layer does not overlap the connection conductors, wherein the second upper redistribution layer overlaps the connection conductors, wherein the first and second upper redistribution layers form a second step with a height substantially equal to or smaller than that of the first step.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a Continuation of U.S. patent application Ser. No. 17/741,581, filed on May 11, 2022, which claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2021-0120816, filed on Sep. 10, 2021 in the Korean Intellectual Property Office, the disclosures of which are incorporated by reference herein in their entirety.

TECHNICAL FIELD

[0002] The present inventive concept relates to a semiconductor package, and more particularly, to a semiconductor package including an encapsulant.

DISCUSSION OF THE RELATED ART

[0003] In accordance with the trend for high performance and high miniaturization of electronic devices, a package-on-package (POP) technology has been under development. Package-on-package technology typically includes semiconductor packages, which include different semiconductor chips, vertically stacked on each other. To implement a package-on-package of a semiconductor package including a high-performance semiconductor chip, it may be desirable to form an upper redistribution layer of the semiconductor package to have a fine pitch.

SUMMARY

[0004] According to an example embodiment of the present inventive concept, a semiconductor package includes: a lower redistribution structure including a lower insulating layer and a lower redistribution layer disposed on the lower insulating layer; a semiconductor chip disposed on the lower redistribution structure, and including a connection pad electrically connected to the lower redistribution layer; connection conductors disposed around the semiconductor chip on the lower redistribution structure, and electrically connected to the lower redistribution layer; an encapsulant disposed on at least a portion of the semiconductor chip and the connection conductors; and an upper redistribution structure including an upper insulating layer and upper redistribution layers, wherein the upper insulating layer is disposed on the encapsulant, wherein the upper redistribution layers are disposed on the upper insulating layer and are electrically connected to the connection conductors, wherein an upper surface of each of the connection conductors and an upper surface of the encapsulant form a first step in a first direction, substantially perpendicular to an upper surface of the lower redistribution structure, wherein the upper redistribution layers include a first upper redistribution layer and a second upper redistribution layer, wherein the first upper redistribution layer does not overlap the connection conductors, wherein the second upper redistribution layer

overlaps the connection conductors, in the first direction, wherein a lower surface of the first upper redistribution layer and a lower surface of the second upper redistribution layer form a second step, and wherein a height of the second step is substantially equal to or smaller than a height of the first step in the first direction.

[0005] According to an example embodiment of the present inventive concept, a semiconductor package includes: a lower redistribution structure including a lower redistribution layer; a semiconductor chip disposed on the lower redistribution structure, and including a connection pad electrically connected to the lower redistribution layer; connection conductors disposed adjacent to the semiconductor chip on the lower redistribution structure, and electrically connected to the lower redistribution layer; an encapsulant covering at least a portion of the semiconductor chip and the connection conductors; and an upper redistribution structure including an upper insulating layer and upper redistribution layers, wherein the upper insulating layer is disposed on an upper surface of the encapsulant and upper surfaces of the connection conductors, wherein the upper redistribution layers are disposed on the upper insulating layer and electrically connected to the connection conductors, wherein the upper surface of the encapsulant is disposed on a level lower than that of the upper surfaces of the connection conductors, and wherein a first portion of the upper redistribution layers disposed on the upper surface of the encapsulant are disposed on a level lower than that of a second portion of the upper redistribution layers disposed on an upper surface of a corresponding connection conductor of the connection conductors.

[0006] According to an example embodiment of the present inventive concept, a semiconductor package includes: a lower redistribution structure including a lower redistribution layer; a semiconductor chip disposed on the lower redistribution structure, and including a connection pad electrically connected to the lower redistribution layer; connection conductors disposed around the semiconductor chip on the lower redistribution structure, and electrically connected to the lower redistribution layer; an encapsulant disposed on at least portions of side surfaces of the connection conductors, and having an upper surface including a plurality of grooves; and an upper redistribution structure including an upper insulating layer and an upper redistribution layer, wherein the upper insulating layer is disposed on the encapsulant and the connection conductors, and the upper redistribution layer is disposed on the upper insulating layer and electrically connected to the connection conductors, and wherein there is a height difference between the upper surface of the encapsulant and an upper surface of a first connection conductor of the connection conductors.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0007] The above and other features of the present inventive concept will become more apparent by describing in detail example embodiments thereof, with reference to the accompanying drawings, in which:

[0008] FIG. 1A is a cross-sectional view illustrating a semiconductor package according to an example embodiment of the present inventive concept, FIG. 1B is a partially enlarged view illustrating region 'A' of FIG. 1A, and FIG. 1C is a plan view illustrating a cross-section taken along line I-I' of FIG. 1A;

[0009] FIGS. 2A, 2B, 2C, 2D, 2E and 2F are cross-sectional views illustrating a manufacturing process of the semiconductor package of FIG. 1A according to an example embodiment of the present inventive concept;

[0010] FIG. 3A is a cross-sectional view illustrating a semiconductor package according to an example embodiment of the present inventive concept, and FIG. 3B is a plan view illustrating a cross-section taken along line II-II' of FIG. 3A; and

[0011] FIG. 4 is a cross-sectional view illustrating a semiconductor package according to an example embodiment of the present inventive concept.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0012] Hereinafter, example embodiments of the present inventive concept will be described with reference to the accompanying drawings.

[0013] FIG. 1A is a cross-sectional view illustrating a semiconductor package according to an example embodiment of the present inventive concept. FIG. 1B is a partially enlarged view illustrating region 'A' of FIG. 1A, and FIG. 1C is a plan view illustrating a cross-section taken along line I-I' of FIG. 1A.

[0014] Referring to FIGS. 1A to 1C, a semiconductor package **100A** according to an example embodiment of the present inventive concept may include a lower redistribution structure **110**, a semiconductor chip **120**, a connection structure **130**, an encapsulant **140**, and an upper redistribution structure **150**. In the present inventive concept, by processing the encapsulant **140** using a laser, a step is formed between an upper surface of the encapsulant **140** and the connection structure **130** (or, e.g., an upper surface of a connection conductor **134**), and at substantially the same time, surface roughness (Ra) is formed in the upper surface of the encapsulant **140**. As a result, according to an example embodiment of the present inventive concept, adhesion of the upper insulating layer **151** may be increased and undulation of the upper insulating layer **151** may be minimized, so that the upper redistribution layer **152** and the upper redistribution via **153** may be formed to have a fine pitch. In addition, since a portion of the encapsulant **140** remains on at least a portion of the upper surface of the connection structure **130**, damage to the connection structure **130** may be prevented during a processing process of the encapsulant **140**.

[0015] The lower redistribution structure **110** is a support substrate on which the semiconductor chip **120** is mounted, and may include a lower insulating layer **111** and lower redistribution conductors **112** and **113** redistributing a connection pad **121** of the semiconductor chip **120**. The lower redistribution conductors **112** and **113** may include a lower redistribution layer **112** and a lower redistribution via **113**. The lower redistribution layer **112** is disposed on the lower insulating layer **111**, and the lower redistribution via **113** penetrates through the lower insulating layer **111**. The lower redistribution structure **110** may include a bigger or smaller number of lower insulating layers **111**, lower redistribution layers **112**, and lower redistribution vias **113** than what is illustrated in the drawings.

[0016] The lower insulating layer **111** may include an insulating resin. For example, the insulating resin may include a thermosetting resin such as an epoxy resin, a thermoplastic resin such as a polyimide, or a resin impregnated with inorganic fillers or/and glass fibers (Glass Fiber, Glass Cloth, Glass Fabric) in these resins, for example, prepreg, Ajinomoto Build-up Film (ABF), FR-4, Bismaleimide Triazine (BT). For example, the lower insulating layer **111** may include a photosensitive resin such as a Photoimageable Dielectric (PID). The lower insulating layer **111** may include a plurality of lower insulating layers **111** stacked in a vertical direction (e.g., Z-axis direction). For example, depending on the process, a boundary between the lower insulating layers **111** of different levels may be unclear or not clearly visible. In addition, for convenience of explanation, only three lower insulating layers **111** are illustrated in the drawings, but example embodiments of the present inventive concept are not limited thereto. According to an example embodiment of the present inventive concept, the lower redistribution structure **110** may also include four or more lower insulating layers **111**.

[0017] The lower redistribution layer **112** may include, for example, copper (Cu), aluminum (Al), silver (Ag), tin (Sn), gold (Au), nickel (Ni), lead (Pb), titanium (Ti), or a metal material including alloys thereof. The lower redistribution layer **112** may include a ground (GND) pattern, a power (PWR) pattern, and a signal(S) pattern. The signal(S) pattern may provide a path through which various signals except for the ground (GND) pattern and the power (PWR) pattern, are transmitted/received. For example, a data signal or the like may be transmitted/received through the

path. The lower redistribution layer **112** may include a plurality of lower redistribution layers respectively disposed below the plurality of lower insulating layers **111**. For example, the plurality of lower redistribution layers **112** and the plurality of lower insulating layers **111** may be alternately stacked on each other. The plurality of lower redistribution layers **112** may be electrically connected to each other through a lower redistribution via **113**. For example, a lowermost redistribution layer **112**, in contact with a connection bump **170**, may be formed to have a greater thickness than the redistribution layers disposed there above. The number of layers of the lower redistribution layers **112** may be determined according to the number of layers of the lower insulating layers **111**, and may include more or fewer layers than shown in the drawings.

[0018] The lower redistribution via **113** may be electrically connected to the lower redistribution layer **112**, and may include a signal via, a ground via, and a power via. The lower redistribution via **113** may include, for example, copper (Cu), aluminum (Al), silver (Ag), tin (Sn), gold (Au), nickel (Ni), lead (Pb), titanium (Ti), or a metal material including alloys thereof. The lower redistribution via **113** may have a form of a via filled with a metal material. For example, the lower redistribution via **113** may be a via hole filled with a metal material or a conformal via in which a metal material is formed along an inner wall of the via hole of the conformal via. The lower redistribution via **113** may be integrated with the lower redistribution layer **112**, but the present inventive concept is not limited thereto.

[0019] The semiconductor chip **120** may be disposed on the lower redistribution structure **110**, and may include a connection pad **121** electrically connected to the lower redistribution layer **112**. The semiconductor chip **120** may be an integrated circuit (IC) in a bare state in which a separate bump or a wiring layer is not formed, but an example embodiment thereof is not limited thereto, and for example, the semiconductor chip **120** may be a packaged-type integrated circuit. Accordingly, the connection pad **121** may be understood as a pad of a bare chip (e.g., an aluminum (Al) pad) or a bump formed on the pad. The integrated circuit may be formed based on an active wafer. The semiconductor chip **120** may include, for example, silicon (Si), germanium (Ge), or gallium arsenide (GaAs), and various types of integrated circuits may be formed therein. The integrated circuit may be a processor chip such as a central processor (e.g., CPU), a graphics processor (e.g., GPU), a field programmable gate array (FPGA), an application processor (AP), a digital signal processor, a cryptographic processor, a microprocessor, and the like, but the present inventive concept is not limited thereto, and for example, the integrated circuit may be a memory chip such as a logic chip such as a central processor (CPU), a graphics processor (GPU), a field programmable gate array (FPGA), a digital signal processor, an encryption processor, a microprocessor, a microcontroller, an analog-digital converter, an application-specific IC (ASIC), or a memory chip such as a volatile memory (e.g., DRAM), a non-volatile memory (e.g., ROM and a flash memory). The connection pad **121** may electrically connect the integrated circuit of the semiconductor chip **120** with other components.

[0020] The connection structure **130** may be disposed on the lower redistribution structure **110** and may provide an electrical path between the lower redistribution layer **112** and the upper redistribution layer **152** around the semiconductor chip **120** by connecting the lower redistribution layer **112** and the upper redistribution layer **152** to each other. The connection structure **130** may include an insulating layer **131** and a connection conductor **134**. The connection conductor **134** extends in a vertical direction (e.g., Z-axis direction) and penetrates the insulating layer **131** to electrically connect the lower redistribution layer **112** and the upper redistribution layer **152** to each other. The connection conductor **134** may include a wiring layer **132** and a wiring via **133**. The wiring layer **132** may be disposed on upper and lower surfaces of the insulating layer **131**, and the wiring via **133** may penetrate through the insulating layer **131**. The connection structure **130** may increase rigidity of the semiconductor package **100A** according to a material of the insulating layer **131**, and may secure thickness uniformity of the encapsulant **140**. The connection structure **130** may have a cavity **130CV** in which the semiconductor chip **120** is disposed.

[0021] The insulating layer **131** may be, for example, an insulating resin. For example, the insulating resin may include a thermosetting resin such as an epoxy resin, a thermoplastic resin such as polyimide, or a resin impregnated with inorganic fillers, for example, ABF, or the like may be used. In addition, a material, in which the resins described above are impregnated with glass fibers, together with the inorganic fillers, for example, prepreg or the like, may be used. According to an example embodiment of the present inventive concept, the connection structure **130** may include only the connection conductor **134** in a form of a post without for the insulating layer **131**. [0022] The wiring layer **132** may provide an upper/lower electrical connection path of the package together with the wiring via **133**, and may serve to redistribute the connection pad **121**. The wiring layer **132** may include, for example, copper (Cu), aluminum (Al), silver (Ag), tin (Sn), gold (Au), nickel (Ni), lead (Pb), titanium (Ti), or a metal material of an alloy thereof. The wiring layer **132**, for example, may include a ground pattern, a power pattern, a signal pattern, etc. according to the design of the corresponding layer. In addition, the wiring layer **132** may be formed by a plating process such as SAP, MSAP, or the like, and may be comprised of a seed layer and a conductor layer. For example, a thickness of each of the wiring layers **132** may be greater than a thickness of each of the lower redistribution layers **112**. The connection structure **130** may have a thickness, greater than or equal to that of the semiconductor chip **120**, and a material of the insulating layer **111** may be selected as prepreg to maintain rigidity, whereas the lower redistribution structure **110** may include a fine circuit and a high-density design. In addition, since a photosensitive resin (e.g., PID) is selected as the material of the insulating layer **111**, the thickness of the wiring layer **132** may be relatively thick compared to the thickness of the lower redistribution layer **112**.

[0023] The wiring via **133** electrically connects the wiring layers **132** formed in different layers to each other, and as a result, an electrical path may be formed in the connection structure **110**. A material for forming the wiring via **133** may be, for example, copper (Cu), aluminum (Al), silver (Ag), tin (Sn), gold (Au), nickel (Ni), lead (Pb), titanium (Ti), a metal material including alloys thereof, or the like. The wiring via **133** may include a signal via, a power via, a ground via, and the like, and the power via and the ground via may be the same via. The wiring via **133** may have a filled-type via or a conformal-type via. The filled-type via may be filled with a metal material, and the conformal-type via may include a metal material that is formed along an inner surface of a via hole of the conformal-type via. For example, each of the wiring vias **133** may have a tapered shape. For example, the wiring via **133** may also be formed by a plating process, and may include a seed layer and a conductor layer. For example, the wiring via **133** may have a shape integrated with the wiring layer **132**. However, the present inventive concept is not limited thereto. For example, the wiring via **133** may be a different material from that of the wiring layer **132**.

[0024] According to an example embodiment of the present inventive concept, since the encapsulant **140** is formed to cover the upper surface of the connection structure **130**, undulation of the upper redistribution structure **150** may be minimized. For example, the encapsulant **140** may cover at least a portion of an upper surface of the insulating layer **131** and at least a portion of a side surface each of the connection conductors **134** (or, e.g., of the wiring layer **132**) protruding above the upper surface of the insulating layer **131**. For example, the encapsulant **140** may expose at least a portion of an upper surface of the wiring layer **132**. In this case, a thickness of the encapsulant **140** covering the upper surface of the connection structure **130** (or, e.g., the insulating layer **131**) may be in a range about 1 μm or more, for example, about 1 μm to about 10 μm , about 2 μm to about 9 μm , about 3 μm to about 9 μm , about 4 μm to about 8 μm . When the thickness of the encapsulant **140** is less than about 1 μm , the connection structure **130** may be damaged during a processing process of the encapsulant **140** (refer to FIG. 2E).

[0025] The encapsulant **140** may encapsulate at least a portion of the semiconductor chip **120** and the connection structure **130** (or the connection conductors **134**) on the lower redistribution structure **110**. For example, the encapsulant may cover at least a portion of the semiconductor chip **120** and the connection structure **130**. The encapsulant **140** may be formed of an insulating resin

(e.g., ABF) in which an inorganic filler is included in an insulating resin such as an epoxy resin. According to example embodiment of the present inventive concept, an upper surface **140S** of the encapsulant **140** may have a step from an upper surface **134S** of the connection conductors **134**, and may include a plurality of grooves. For example, the upper surface **140S** of the encapsulant **140** may be uneven. The upper surface **140S** of the encapsulant **140** may be disposed on a level lower than that of the upper surface **134S** of the connection conductors **134**. The plurality of grooves may provide surface roughness (Ra) on an upper surface **140S** of the encapsulant **140**, and may increase adhesion between the encapsulant **140** and the upper insulating layer **151**. In one example, the upper surface **140S** of the encapsulant **140** may have surface roughness in a range of about 1 μm or more, for example, about 1 μm to about 7 μm , about 2 μm to about 6 μm , or about 3 μm to about 5 μm .

[0026] The upper redistribution structure **150** may include an upper insulating layer **151**, the upper redistribution layers **152**, and the upper redistribution via **153**. The insulating layer **151** may be disposed on the encapsulant **140** and the connection conductors **134**. The upper redistribution layers **152** may be disposed on the upper insulating layer **151** and electrically connected to the connection conductors **134**, and the upper redistribution via **153** may penetrate through the upper insulating layer **151** to connect the upper redistribution layers **152** to the connection conductors **134**. Since the upper insulating layer **151**, the upper redistribution layer **152**, and the upper redistribution via **153** have the same or similar characteristics (e.g., configurations, arrangements and components) to the lower insulating layer **111**, the lower redistribution layer **112**, and the lower redistribution via **113** described above, overlapping descriptions thereof will be omitted. The upper redistribution layer **152** and the upper redistribution via **153** may include a seed layer **154** disposed between the upper insulating layer **151** and the upper redistribution layer **152** and between the upper insulating layer **151** and the upper redistribution via **153**. For example, seed layer may be formed of a metal such as titanium (Ti) or copper (Cu), or an alloy thereof.

[0027] The upper insulating layer **151** may be disposed on an upper surface **140S** of the encapsulant **140**. For example, the upper insulating layer **151** may be in direct contact with an upper surface **140S** of the encapsulant **140**, and may include an insulating resin different from that of the encapsulant **140**. The upper insulating layer **151** may include a material suitable for forming the upper redistribution layer **152** and the upper redistribution vias **153** to have a fine pitch, for example, a photosensitive resin such as PID. In addition, the encapsulant **140** may include a non-photosensitive resin (e.g., ABF) containing an inorganic filler to protect the semiconductor chip **120** and ensure durability of the semiconductor package **100A**. Since the upper surface **140S** of the encapsulant **140** has surface roughness (Ra) formed by laser processing, adhesion of the upper insulating layer **151** may be increased. An upper surface of the upper insulating layer **151** may have a recess portion **151R** that is recessed corresponding to a step between an upper surface **134S** of the connection conductors **134** and an upper surface **140S** of the encapsulant **140**. For example, the recess portion **151R** may overlap the step formed between the upper surface **134S** of the connection conductors **134** and the upper surface **140S** of the encapsulant **140**. For example, the recess portion **151R** may be disposed between the connection conductors **134**. As another example, the recess portion **151R** may overlap the grooves formed in the upper surface **140S** of the encapsulant **140**. In an example embodiment of the present inventive concept, since an encapsulant **140** covering an upper surface of the connection structure **130** is disposed between the connection conductors **134**, a depth of the recess portion **151R** may be minimized. The depth of the recess portion **151R** may affect a fine pitch of the upper redistribution layer **152** and the upper redistribution via **153**, and according to the present inventive concept, a maximum depth of the recess portion **151R** may be about 1 μm or less.

[0028] The upper redistribution layers **152** may be disposed on an upper surface of the upper insulating layer **151**, and may be electrically connected to the connection conductor **134** through the upper redistribution via **153**. For example, at least a portion (e.g., **152a**) of the upper

redistribution layers **152** disposed above the upper surface **140S** of the encapsulant **140** may be disposed on a level lower than that of the other portion (e.g., **152b**) of the upper redistribution layers **152** disposed above the upper surface **134S** of the connection conductors **134**. For example, the portion (e.g., **152a**) of the upper redistribution layers **152**, which may be disposed in the recess portion **151R** of the upper insulating layer **151**, may be disposed on a level lower than that of the other portion (e.g., **152b**) of the upper redistribution layers **152** overlapping the upper surface **134S** of the connection conductors **134**. Here, a lower surface of at least a portion (e.g., **152a**) of the upper redistribution layers **152** may have a step **d2** of about 1 μm or less with a lower surface of the other portion (e.g., **152b**) of the upper redistribution layers **152**. For example, the lower surface of the portion (e.g., **152a**) of the upper redistribution layers **152** be disposed lower than the lower surface of the other portion (e.g., **152b**) of the upper redistribution layers **152** by about 1 μm . For example, the upper redistribution layers **152** may have a step according to the shapes of the structures thereunder, for example, the connection conductors **134** and the encapsulant **140**, and according to the present inventive concept, since a step thickness difference between the upper redistribution layer **152** may be minimized, the upper redistribution layers **152** having a finer pitch may be implemented. For example, the connection conductors **134** may protrude further than the upper surface **140S** of the encapsulant **140**. For example, the upper surface **134S** of each of the connection conductors **134** and the upper surface **140S** of the encapsulant **140** may have a first step (**d1**) in a first direction (e.g., the Z-axis direction), substantially perpendicular to an upper surface of the lower redistribution structure **110**. As another example, the upper redistribution layers **152** may include a first upper redistribution layer **152a** and a second upper redistribution layer **152b**. The first upper redistribution layer **152a** does not overlap the connection conductors **134** in the first direction (e.g., the Z-axis direction), and the second upper redistribution layer **152b** overlap the connection conductors **134**. As an additional example, a lower surface of the first upper redistribution layer **152a** and a lower surface of the second upper redistribution layer **152b** may have a second step **d2**, substantially equal to or smaller than the first step **d1**. Here, a height of the first step **d1** may be in a range of about 5 μm or less, for example, about 0.0001 μm to about 5 μm , about 0.0001 μm to about 4 μm , or about 0.0001 μm to about 3 μm . In addition, a height of the second step **d2** may be in a range of about 1 μm or less, for example, about 0.001 μm to about 1 μm , or about 0.0001 μm to about 1 μm . The lower surface of the first upper redistribution layer **152a** and the lower surface of the second upper redistribution layer **152b** may be interpreted as a lower surface of the seed layer **154**, respectively, and it can be interpreted that a maximum depth of the recess portion **151R** is substantially equal to the height of the second step **d2**. In addition, in FIG. 1C, the first upper redistribution layer **152a** and the second upper redistribution layer **152b** are illustrated as a plurality of redistribution layers, spaced apart from each other, but this is for convenience of description, and the present inventive concept is not limited thereto and the first upper redistribution layer **152a** and the second upper redistribution layer **152b** are not different redistribution layers, electrically and physically independent from each other. For example, each of the first upper redistribution layer **152a** and the second upper redistribution layer **152b** may be partial regions of one upper redistribution layer **152** extending above the connection conductor **134** and the encapsulant **140**.

[0029] According to an example embodiment of the present inventive concept, the semiconductor package **100A** may further include a first solder resist layer **SR1** and a second solder resist layer **SR2**. The first solder resist layer **SR1** may be disposed below the lower redistribution structure **110**, and the second solder resist layer **SR2** disposed above the upper redistribution structure **150**. The first and second solder resist layers **SR1** and **SR2** may protect the lower redistribution layer **112** and the upper redistribution layer **152** from physical and chemical damage. The first and second solder resist layers **SR1** and **SR2** may include an insulating material, and may be formed using, for example, prepreg, ABF, FR-4, BT, or a photo solder resist (PSR).

[0030] In addition, a UBM structure **160** and/or a connection bump **170** electrically connected to

the lower redistribution layer **112** may be disposed below the lower redistribution structure **110**. The UBM structure **160** may increase connection reliability of the connection bump **170**, and as a result, board level reliability of the semiconductor package **100A** may be increased. The connection bump **170** may be disposed below the UBM structure **160**, and may have, for example, a spherical or ball shape made of a low-melting-point metal, for example, tin (Sn), indium (In), bismuth (Bi), antimony (Sb), copper (Cu), silver (Ag), zinc (Zn), lead (Pb), an alloy containing the same (e.g., Sn-Ag-Cu), or the like.

[0031] FIGS. 2A to 2F are cross-sectional views illustrating a manufacturing process of the semiconductor package **100A** of FIG. 1A according to an example embodiment of the present inventive concept.

[0032] Referring to FIG. 2A, a semiconductor chip **120** and a connection structure **130** (or a connection conductor **134**) may be disposed on a first carrier **10**. The first carrier **10** is a temporary carrier having adhesive properties, and may support the semiconductor chip **120** and the connection structure **130**. For example, the semiconductor chip **120** may be disposed in a cavity **130CV** at least partially surrounded by the connection structure **130**. The connection structure **130** may include a connection conductor **134** providing an electrical path in a vertical direction (e.g., the Z-axis direction) along a side surface of the semiconductor chip **120**.

[0033] Referring to FIG. 2B, a preliminary encapsulant **140p** for sealing the semiconductor chip **120** and the connection structure **130** may be formed on the first carrier **10**. The preliminary encapsulant **140p** may be formed by, for example, transferring a film-type insulating resin such as ABF onto the first carrier **10**. The preliminary encapsulant **140p** may be formed to cover an upper surface of the semiconductor chip **120** and an upper surface of the connection structure **130**. Thereafter, the first carrier **10** may be removed, and the semiconductor chip **120** and the connection structure **130** sealed by the preliminary encapsulant **140p** may be attached to a second carrier ('**20**' in FIG. 2C) so that the connection pad **121** of the semiconductor chip **120** is exposed.

[0034] Referring to FIG. 2C, a lower redistribution structure **110** may be formed on the semiconductor chip **120** and the connection structure **130**. The lower redistribution structure **110** may include a lower insulating layer **111**, a lower redistribution layer **112**, and a lower redistribution via **113**. The lower insulating layer **111** may be formed by coating and curing a photosensitive material, for example, PID. Next, an exposure process and a development process by using a photomask may be performed to form via holes penetrating through the lower insulating layer **111**. Next, a lower redistribution via **113** and a lower redistribution layer **112** may be formed using a deposition process, a plating process, an etching process, or the like. By repeating the above-described process, the lower redistribution structure **110** including the plurality of insulating layers **111** and the plurality of lower redistribution layers **112** may be formed. In an example embodiment of the present inventive concept, a first solder resist layer **SR1** and a UBM structure **160** may be formed on the lower redistribution structure **110**.

[0035] Referring to FIG. 2D, thereafter, the second carrier **20** may be removed, and a lower redistribution structure **110** may be attached to a third carrier **30** such that a preliminary encapsulant **140p** faces an upward direction. The third carrier **30** may include an adhesive layer **31** and a core layer **32**. The adhesive layer **31** may be attached to the lower redistribution structure **110**, and the core layer **32** may support the adhesive layer **31**. The adhesive layer **31** may be made of an adhesive polymer and a release layer. The release layer may be for peeling the core layer **32**, and may be interposed between the adhesive layer **31** and the core layer **32**.

[0036] Referring to FIG. 2E, the preliminary encapsulant **140p** of FIG. 2D may be processed to form an encapsulant **140**. The encapsulant **140** may be formed by removing a portion of the preliminary encapsulant **140p** through laser ablation. Accordingly, an upper surface of the encapsulant **140** may be disposed on a lower level than an upper surface of the connection conductor **134**, and may include a plurality of grooves providing surface roughness (Ra) of a certain (or predetermined) level or higher. Since the encapsulant **140** is formed to cover an upper

surface of the connection structure **130**, damages to the connection structure **130** by laser processing may be prevented. In addition, since the upper surface of the encapsulant **140** has surface roughness (Ra) formed from laser processing, it is possible to secure adhesion with an upper insulating layer ('**151**' in FIG. 5F) that is to be formed later. Moreover, the encapsulant **140** may fill a space between the connection conductors **134**, thereby minimizing a depth of a recess portion formed in the upper insulating layer ('**151**' in FIG. 5F). Laser processing can effectively remove the preliminary encapsulant **140p** and prevent damage to the connection structure **130** by controlling conditions (e.g., power, etc.) of a laser beam. For example, laser processing may be performed using a continuous wave (CW) laser or a pulsed laser, for example, a femtosecond laser. [0037] Referring to FIG. 2F, an upper redistribution structure **150** may be formed on the encapsulant **140** and the connection conductors **134**. The upper redistribution structure **150** may include an upper insulating layer **151**, upper redistribution layers **152**, and upper redistribution vias **153**. The upper insulating layer **151**, the upper redistribution layer **152**, and the upper redistribution via **153** may be formed in a similar manner to that of the lower insulating layer **111**, the lower redistribution layer **112**, and the lower redistribution via **113** described with reference to FIG. 2C. According to the present inventive concept, since the undulation of the upper insulating layer **151** and the step thickness difference between the upper redistribution layers **152** are minimized, it is possible to form the upper redistribution layers **152** and the upper redistribution vias **153** with a fine pitch. In an example embodiment of the present inventive, a second solder resist layer SR2 covering at least a portion of the upper redistribution layers **152** may be formed on the upper redistribution structure **150**.

[0038] FIG. 3A is a cross-sectional view illustrating a semiconductor package **100B** according to an example embodiment of the present inventive concept, and FIG. 3B is a plan view illustrating a cross-section taken along line II-II' of FIG. 3A.

[0039] Referring to FIGS. 3A and 3B a semiconductor package **100B** according to an example embodiment of the present inventive concept may have same or similar characteristics (e.g., configurations, arrangements, and components) as the semiconductor package **100A** illustrated in FIGS. 1A to 1C, except that the semiconductor package **100B** includes a plurality of semiconductor chips **120a** and **120b**. For example, it may be understood that the semiconductor package **100B** according to an example embodiment of the present inventive concept has a structure similar to that illustrated in FIG. 1C. The plurality of semiconductor chips **120a** and **120b** may be disposed on the lower redistribution structure **110** and may be spaced apart from each other in a horizontal direction. In addition, the plurality of semiconductor chips **120a** and **120b** may include different types of semiconductor chips. For example, a 1-1 semiconductor chip **120a** may include an AP chip, and a 1-2 semiconductor chip **120b** may include a power management integrated circuit (PMIC) chip.

[0040] The connection structure **130** may include a plurality of cavities **130CV1** and **130CV2** accommodating the plurality of semiconductor chips **120a** and **120b**, respectively. For example, the connection structure **130** may include a first cavity **130CV1**, in which the 1-1 semiconductor chip **120a** is disposed, and a second cavity **130CV2**, in which the 1-2 semiconductor chip **120b** is disposed. According to an example embodiment of the present inventive concept, the first cavity **130CV1** and the second cavity **130CV2** may be formed as a single cavity connected to each other.

[0041] FIG. 4 is a cross-sectional view illustrating a semiconductor package **1000** according to an example embodiment of the present inventive concept.

[0042] Referring to FIG. 4, the semiconductor package **1000** according to an example embodiment of the present inventive concept may have a package-on-package structure including a first package **100** and a second package **200**. The first package **100** may have the same or similar characteristics to the semiconductor packages **100A** and **100B** described with reference to FIGS. 1A to 3.

[0043] The second package **200** may include a second redistribution substrate **210**, a second semiconductor chip **220**, and a second encapsulant **230**. The second redistribution substrate **210**

may include a lower pad **211** and an upper pad **212** electrically connected to the outside, and the lower pad **211** and the upper pad **212** are respectively disposed on a lower surface and an upper surface of the second redistribution substrate **210**. The upper pad **212** and the lower pad **211** may be electrically connected to each other. In addition, in the second redistribution substrate **210**, a redistribution circuit **213** may connect the lower pad **211** and the upper pad **212** to each other. [0044] The second semiconductor chip **220** may be mounted on the second redistribution substrate **210** by wire bonding or flip-chip bonding. For example, the plurality of second semiconductor chips **220** may be vertically stacked on the redistribution substrate **210**, and electrically connected to the upper pad **212** of the redistribution substrate **210** by a bonding wire WB. In one example, the second semiconductor chip **220** may include a memory chip, and the first semiconductor chip **120** may include an AP chip.

[0045] The second encapsulant **230** may include the same or similar material to the encapsulant **140** of the first package **100**. The second package **200** may be physically and electrically connected to the first package **100** by a metal bump **214**. The metal bump **214** may be electrically connected to the redistribution circuit **213** inside the redistribution substrate **210** through the lower pad **211** on the lower surface of the redistribution substrate **210**. The metal bump **214** may be formed of a low-melting-point metal, for example, tin (Sn) or an alloy including tin (Sn).

[0046] As set forth above, according to example embodiment of the present inventive concept, a semiconductor package including an upper redistribution structure having a fine pitch may be provided.

[0047] It is to be understood that spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, in the example, terms “below” and “beneath” may encompass both an orientation of above, below and beneath. The device may be otherwise oriented (e.g., rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may be interpreted accordingly.

[0048] The meaning of a “connection” of a component to another component in the description includes an indirect connection through an adhesive layer as well as a direct connection between two components. It will be understood that, although the terms “first”, “second”, etc. may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms are only used to distinguish one element, component, region, layer or section from another element, component, region, layer or section. Thus, a first element, component, region, layer or section discussed below could be termed a second element, component, region, layer or section without departing from the spirit and scope of the present invention.

[0049] It is to be understood that example embodiments provided herein are considered to be able to be implemented by being combined in whole or in part one with one another. For example, one element described in a particular example embodiment, even if it is not described in another example embodiment, may be understood as a description related to another example embodiment, unless an opposite or contradictory description is provided therein.

[0050] It is to be understood that singular forms include plural forms unless the context clearly indicates the contrary.

[0051] While the present inventive concept has been particularly shown and described with reference to example embodiments thereof, it will be apparent to those of ordinary skill in the art that various changes in form and detail may be made thereto without departing from the spirit and scope of the present inventive concept

Claims

- 1.** A method of manufacturing a semiconductor package, comprising: disposing a semiconductor chip and a plurality of connection conductors on a first carrier, the semiconductor chip including a first surface facing the first carrier, a second surface opposite to the first surface, and a connection pad on the first surface; forming a preliminary encapsulant on the second surface of the semiconductor chip, the preliminary encapsulant sealing the plurality of connection conductors and the semiconductor chip; attaching a second carrier on the preliminary encapsulant and removing the first carrier; forming a first redistribution structure on the first surface of the semiconductor chip, the first redistribution structure including a first insulating layer, and a first redistribution layer disposed on the first insulating layer and electrically connected to the connection pad and the plurality of connection conductors; attaching a third carrier on the first redistribution structure and removing the second carrier; forming an encapsulant by removing a portion of the preliminary encapsulant through laser ablation, wherein an upper surface of the encapsulant has a first step with respect to an upper surface of each of the plurality of connection conductors in a vertical direction; and forming a second redistribution structure on the encapsulant, the second redistribution structure including a second insulating layer covering at least portions of each of the plurality of connection conductors, and a second redistribution layer disposed on the second insulating layer and electrically connected to the plurality of connection conductors, wherein the second redistribution layer includes a first upper redistribution pattern and a second upper redistribution pattern, wherein a second step between a lower surface of the first upper redistribution pattern and a lower surface of the second upper redistribution pattern is equal to or smaller than the first step in the vertical direction.
- 2.** The method of manufacturing the semiconductor package of claim 1, wherein the plurality of connection conductors is disposed around the semiconductor chip.
- 3.** The method of manufacturing the semiconductor package of claim 2, wherein the first upper redistribution pattern does not overlap the plurality of connection conductors in the vertical direction, and the second upper redistribution pattern overlaps the plurality of connection conductors in the vertical direction.
- 4.** The method of manufacturing the semiconductor package of claim 1, wherein the upper surface of each of the plurality of connection conductors is above the upper surface of the encapsulant in the vertical direction.
- 5.** The method of manufacturing the semiconductor package of claim 1, wherein the upper surface of the encapsulant has a surface roughness of 1 μm or more.
- 6.** The method of manufacturing the semiconductor package of claim 1, wherein the second insulating layer has a recess portion.
- 7.** The method of manufacturing the semiconductor package of claim 6, wherein the first upper redistribution pattern is disposed in the recess portion.
- 8.** The method of manufacturing the semiconductor package of claim 6, wherein the recess portion is disposed between the plurality of connection conductors.
- 9.** The method of manufacturing the semiconductor package of claim 6, wherein a maximum depth of the recess portion is equal to the second step.
- 10.** The method of manufacturing the semiconductor package of claim 1, wherein the first step is 5 μm or less, and the second step is 1 μm or less.
- 11.** The method of manufacturing the semiconductor package of claim 1, wherein the second insulating layer includes a photosensitive resin, and the encapsulant includes a non-photosensitive resin.
- 12.** A method of manufacturing a semiconductor package, comprising: disposing a semiconductor chip and a plurality of connection conductors, the semiconductor chip including a first surface

facing the first carrier, a second surface opposite to the first surface, and a connection pad on the first surface; forming a preliminary encapsulant sealing the second surface of the semiconductor chip; forming a first redistribution structure on the first surface of the semiconductor chip, the first redistribution structure including a first redistribution layer electrically connected to the connection pad of the semiconductor chip and the plurality of connection conductors; forming an encapsulant from the preliminary encapsulant by laser ablation, wherein an upper surface of the encapsulant is disposed on a level lower than that of an upper surface of each of the plurality of connection conductors in a vertical direction; and forming a second redistribution structure on the encapsulant, the second redistribution structure including a second redistribution layer electrically connected to the plurality of connection conductors, wherein the second redistribution layer includes a first upper redistribution pattern overlapping a gap between the plurality of connection conductors in the vertical direction, and a second upper redistribution pattern overlapping a corresponding one of the plurality of connection conductors in the vertical direction, wherein a lower surface of the first upper redistribution pattern is disposed on a level lower than that of a lower surface of the second upper redistribution pattern.

13. The method of manufacturing the semiconductor package of claim 12, wherein a distance between the lower surface of the first upper redistribution pattern and the lower surface of the second upper redistribution pattern is 1 μm or less.

14. The method of manufacturing the semiconductor package of claim 12, wherein a distance between the upper surface of the encapsulant and the upper surface of each of the plurality of connection conductors is 5 μm or less.

15. The method of manufacturing the semiconductor package of claim 12, wherein the upper surface of the encapsulant has a plurality of grooves.

16. The method of manufacturing the semiconductor package of claim 15, wherein the upper surface of the encapsulant has a surface roughness of 1 μm or more.

17. The method of manufacturing the semiconductor package of claim 12, wherein the encapsulant includes a non-photosensitive resin impregnated with an inorganic filler.

18. A method of manufacturing a semiconductor package, comprising: disposing a semiconductor chip and a plurality of connection conductors, the semiconductor chip including a first surface facing the first carrier, and a second surface opposite to the first surface; forming a preliminary encapsulant sealing the second surface of the semiconductor chip; forming a first redistribution structure on the first surface of the semiconductor chip, the first redistribution structure including a first redistribution layer electrically connected to the semiconductor chip and the plurality of connection conductors; and forming an encapsulant from the preliminary encapsulant by laser ablation, wherein an upper surface of the encapsulant has a plurality of grooves, and a height difference with respect to an upper surface of each of the plurality of connection conductors.

19. The method of manufacturing the semiconductor package of claim 18, wherein the upper surface of the encapsulant has a surface roughness of 1 μm or more.

20. The method of manufacturing the semiconductor package of claim 18, further comprises, forming a second redistribution structure on the encapsulant, the second redistribution structure including an insulating layer on the plurality of connection conductors and the encapsulant, and a second redistribution layer on the insulating layer.
